Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 1 / 方法 1 [GE05-M01]

Triangle centers are useful because they preserve exact ratios, equal distances and half-angle relations. Choose the appropriate center identity, retain internal/external bisector branches, and connect lengths to area only after the point order is established.

三角形中心可提供精确比例、等距及半角关系。应选择合适的中心恒等式，保留内外角平分线分支，并先明确点序再联系长度与面积。

Angle-bisector ratios persist in nested triangles

角平分线比例在嵌套三角形中仍成立

Procedure

1. For the internal bisector AD, write $BD/DC = AB/AC$ with positive segment lengths.

2. If a second segment crosses the same angle, apply the theorem inside the smaller triangle as well.

3. For a needed length use $AD^2 = AB \cdot AC - BD \cdot DC$, or Stewart's theorem for a general cevian.

4. Restore the signs and triangle inequalities after eliminating or squaring.

通用步骤

1. 对内角平分线 AD，写出正线段比例 $BD/DC = AB/AC$。

2. 若另一条线段横跨同一角，在小三角形内也可使用同一定理。

3. 需要长度时用 $AD^2 = AB \cdot AC - BD \cdot DC$，一般劈线则用斯图尔特定理。

4. 消元或平方后补查符号与三角不等式。

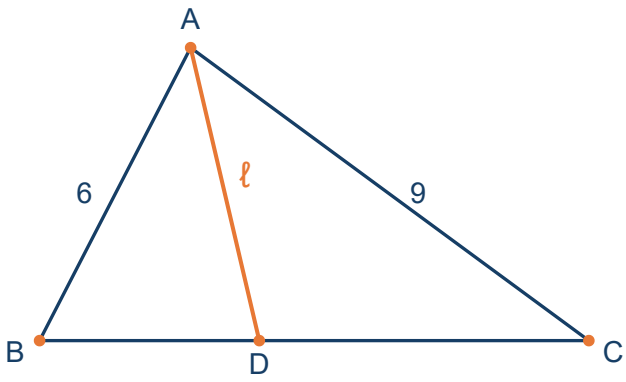
Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 1 / 方法 1 [GE05-M01] (continued / 续)

> Example. A triangle has $AB = 6, AC = 9, BC = 10$. Find the length of the internal bisector from A.

> 例题：三角形满足 $AB = 6, AC = 9, BC = 10$，求从 A 出发的内角平分线长度。



BC = 10 ∠BAD = ∠DAC

Worked solution

1. The ratio $BD/DC = 2/3$ gives $BD = 4, DC = 6$.

2. The bisector identity gives $AD^2 = 6 \cdot 9 - 4 \cdot 6 = 30$.

3. Thus $AD = \sqrt{30}$. The original sides satisfy all strict triangle inequalities, so the internal construction is valid.

例题解析

1. 由 $BD/DC = 2/3$ 得 $BD = 4, DC = 6$。

2. 角平分线恒等式给出 $AD^2 = 6 \cdot 9 - 4 \cdot 6 = 30$。

3. 故 $AD = \sqrt{30}$；原三边满足严格三角不等式，内部构造合法。

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 1 / 方法 1 [GE05-M01] (continued / 续)

An angle bisector is not generally a median.

角平分线一般不是中线。

Stewart's theorem for $BD = m, DC = n, BC = m + n$, $AB = c, AC = b, AD = d$ is $b^2m + c^2n = (m + n)(d^2 + mn)$. Internal-bisector substitution yields the short identity above.

A bisected subangle can create an isosceles auxiliary triangle. Determine its other two angles before assuming the bisector splits the original side in half.

若 $BD = m, DC = n, BC = m + n, AB = c, AC = b, AD = d$，斯图尔特定理为 $b^2m + c^2n = (m + n)(d^2 + mn)$；代入内角平分线比例即可得到上面的简式。

被平分的子角可能产生等腰辅助三角形，应先求另外两角，不能直接假设原边被平分。

Practice / 配套练习: AIME-2001-I-Q04 (WB p. 257), AIME-2002-I-Q10 (WB p. 314), AIME-2009-I-Q05 (WB p. 221), AIME-2010-I-Q15 (WB p. 320), AIME-2010-II-Q15 (WB p. 296), AIME-2012-I-Q12 (WB p. 321), AIME-2026-II-Q10 (WB p. 291)

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 2 / 方法 2 [GE05-M02]

Balance masses to locate cevian intersections

平衡质量定位劈线交点

Procedure

1. Assign positive endpoint masses inversely proportional to the side-division lengths.

2. Scale the assignments so shared vertices carry the same mass.

3. Add endpoint masses when replacing a side by its division point, then read a new ratio inversely.

通用步骤

1. 端点质量按分段长度的反比分配。

2. 统一缩放，使共用顶点质量一致。

3. 将两端质量相加作为分点质量，再按反比读取新的线段比例。

> Example. In triangle ABC, $D \in BC$ has $BD : DC = 2 : 3$, and $E \in CA$ has $CE : EA = 1 : 2$. If $P = AD \cap BE$, find $AP : PD$.

> 例题：三角形中 $BD : DC = 2 : 3, CE : EA = 1 : 2$，且 $P = AD \cap BE$，求 $AP : PD$。

Worked solution

1. Take masses $m_B = 3, m_C = 2$ to balance D. The ratio at E requires $m_A = 1$.

2. The mass at D is $m_B + m_C = 5$.

3. On AD, the balance relation is $AP/PD = m_D/m_A = 5$, so the ratio is $5 : 1$.

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 2 / 方法 2 [GE05-M02] (continued / 续)

例题解析

1. 由 D 取质量 $m_B = 3, m_C = 2$；E 的比例再要求 $m_A = 1$。

2. D 的质量为 $m_B + m_C = 5$。

3. 沿 AD，平衡关系给出 $AP/PD = m_D/m_A = 5$，故为 $5:1$。

> Masses add at a side point; segment lengths do not.

> 分点处相加的是质量，不是线段长度。

Practice / 配套练习: AIME-2002-II-Q13 (WB p. 231), AIME-2011-II-Q04 (WB p. 311)

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 3 / 方法 3 [GE05-M03]

Reflect across a bisector to create an isosceles midpoint

关于角平分线反射构造等腰中点

Procedure

1. Reflection across an internal angle bisector exchanges the two side rays.

2. Reflect a known endpoint onto the other side at the same distance from the vertex.

3. Transfer equal distances to the reflected points, then use perpendicular bisectors, midpoint theorems or parallel-line isosceles triangles.

通用步骤

1. 关于内角平分线反射会交换两条边射线。

2. 把已知端点映到另一边上距顶点相等的位置。

3. 将等距条件转移到反射点，再用垂直平分线、中点定理或平行线等腰三角形。

> Example. Angle A is bisected by line AM. Point B lies on one ray with $AB = 3$. Its reflection is $B\prime$. If M lies on AM and $MB = 4$, find $MB\prime$.

> 例题：直线 AM 平分角 A，一边上 $AB = 3$，将 B 反射得 $B\prime$。若 M 在平分线上且 $MB = 4$，求 $MB\prime$。

Worked solution

1. The reflection fixes A, M and sends B onto the other ray with $AB\prime = 3$.

2. Reflections preserve all distances, so $MB\prime = MB = 4$.

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 3 / 方法 3 [GE05-M03] (continued / 续)

3. Thus M lies on the perpendicular bisector of $BB\prime$, a useful linear condition when several bisectors meet a midpoint.

例题解析

1. 反射固定 A, M，并把 B 映到另一射线上，$AB\prime = 3$。

2. 反射保持距离，故 $MB\prime = MB = 4$。

3. 因此 M 在 $BB\prime$ 的垂直平分线上；多个角平分线通过中点时，这可给出线性条件。

Reflection preserves a ray's distance from the vertex, not its original coordinate.

反射保持到顶点的距离，不保持原坐标。

In a trapezoid, parallel bases often turn an angle-bisector intercept triangle into an isosceles triangle. The second bisector then becomes its median, placing the intersection at a midpoint.

To align coordinates with a bisector, place adjacent vertices at $(c\cos t, c\sin t), (a\cos t, -a\sin t)$. Locate the axis intercept by the bisector theorem before intersecting a perpendicular cut.

梯形中，平行底边常使角平分线截出的三角形成为等腰；另一条平分线于是成为中线，使交点成为中点。

若把平分线作为坐标轴，可将邻接两顶点写为 $(c\cos t, c\sin t), (a\cos t, -a\sin t)$。先用平分线定理定位轴上截点，再求垂直截线。

Practice / 配套练习: AIME-2003-I-Q15 (WB p. 387), AIME-2011-I-Q04 (WB p. 310), AIME-2022-I-Q03 (WB p. 312), AIME-2022-II-Q11 (WB p. 327)

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 4 / 方法 4 [GE05-M04]

Cancel area in inradius, altitude and semiperimeter equations

在内切圆、三角高及半周长式中消去面积

Procedure

1. Write $K = rs$ and $K = ah/2$ for the same triangle.

2. Ratios such as $r/h = a/(2s)$ eliminate area without computing radicals.

3. For equal inradii in adjacent triangles, replace radius equality by area divided by semiperimeter equality.

4. If tangent lengths are used, combine $K = rs$ with Heron and verify the circle is an incircle.

通用步骤

1. 对同一三角形写 $K = rs$ 与 $K = ah/2$。

2. 用 $r/h = a/(2s)$ 等比例消去面积，无须先计算根式。

3. 相邻三角形内切圆相等时，改写为面积除以半周长相等。

4. 若使用切线长，可联立 $K = rs$ 与海伦公式，并确认所用圆确为内切圆。

> Example. A triangle has sides $5, 6, 7$. Through its incenter draw a line parallel to the side of length six. Find the resulting interior parallel chord.

> 例题：三角形三边为 $5, 6, 7$，过内心作平行于六长边的直线，求内部平行截段长。

Worked solution

1. The semiperimeter is nine. If altitude to the six-side is h, then $r/h = 6/(2 \cdot 9) = 1/3$.

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 4 / 方法 4 [GE05-M04] (continued / 续)

2. The upper similar triangle has height $h - r = 2h/3$.

3. Its base chord is therefore $(2/3)6 = 4$.

例题解析

1. 半周长为九。以六长边为底的高为 h，则 $r/h = 6/(2 \cdot 9) = 1/3$。

2. 上方相似小三角形高为 $h - r = 2h/3$。

3. 所以平行底截段长为 $(2/3)6 = 4$。

Use the semiperimeter of the correct subtriangle.

应使用对应子三角形的半周长。

Practice / 配套练习: AIME-2001-I-Q07 (WB p. 224), AIME-2001-II-Q07 (WB p. 313), AIME-2009-I-Q12 (WB p. 319), AIME-2010-I-Q15 (WB p. 320), AIME-2005-I-Q15 (WB p. 294)

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 5 / 方法 5 [GE05-M05]

Tangent lengths are semiperimeter differences

切线长等于半周长之差

Procedure

1. For sides a, b, c and semiperimeter s, the two tangent lengths from A equal $s - a$.

2. Use these values to locate incircle contact points on a common side.

3. Connect a vertex tangent length to the half-angle by $r = (s - a)\tan(A/2)$.

4. When chords or a second tangent meet a triangle side, retain the actual segment order before adding lengths.

通用步骤

1. 三边为 a, b, c、半周长为 s 时，从 A 出发的两切线长为 $s - a$。

2. 利用这些长度定位公共边上的内切圆切点。

3. 由 $r = (s - a)\tan(A/2)$ 将顶点切线长联系到半角。

4. 弦或另一切线与边相交时，先明确线段次序，再作长度相加。

> Example. The incircle of a 5–12–13 right triangle touches the hypotenuse. Find the two hypotenuse pieces.

> 例题：五、十二、十三直角三角形的内切圆与斜边相切，求斜边两分段长。

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 5 / 方法 5 [GE05-M05] (continued / 续)

Worked solution

1. The semiperimeter is fifteen. At the endpoints of the hypotenuse the opposite side lengths are twelve and five.

2. The tangent pieces are therefore $15-12=3$ and $15-5=10$.

3. They sum to thirteen, checking the hypotenuse length. Equal tangents transfer the same values onto the adjacent legs.

例题解析

1. 半周长为十五，斜边两端对应的对边长为十二与五。

2. 所以切线分段为 $15-12=3$ 与 $15-5=10$。

3. 两段和为十三，核实斜边长度；等切线可把相同数值转移至邻接直角边。

Tangent-length addition differs from secant-power multiplication.

切线段相加与割线点幂相乘不可混用。

A circle row along a triangle base is controlled by the same half-angle cotangents as its incircle. Compare two rows to recover the common base coefficient without solving either base angle.

三角形底边圆链由与内切圆相同的两个半角余切控制。比较两种圆链可求共同底边系数，无需分别求两个底角。

Practice / 配套练习: AIME-2007-II-Q09 (WB p. 315), AIME-2018-II-Q14 (WB p. 301), AIME-2022-I-Q11 (WB p. 308), AIME-2024-I-Q08 (WB p. 338)

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 6 / 方法 6 [GE05-M06]

Circumradius converts a side into a sine

外接圆半径把边长转化为正弦

Procedure

1. Use $a = 2R\sin A$ or $R = abc/(4K)$ to connect a center radius with triangle metrics.

2. For an equidistant apex over a triangle plane, its projection is the base circumcenter.

3. Retain the correct half-angle and diagonal choices in a rhombus, and verify the circumcenter location when a filled region is involved.

通用步骤

1. 使用 $a = 2R\sin A$ 或 $R = abc/(4K)$ 联系圆心半径与三角形度量。

2. 到三角形三个顶点等距的空间顶点，其投影为底面外心。

3. 菱形中保留正确半角与对角线选择；涉及填满区域时核实外心位置。

> Example. A triangle has sides $13, 14, 15$. Find its circumradius.

> 例题：三角形三边为 $13, 14, 15$，求外接圆半径。

Worked solution

1. The semiperimeter is twenty-one, so Heron gives $K = \sqrt{21 \cdot 8 \cdot 7 \cdot 6} = 84$.

2. Use $R = abc/(4K) = 13 \cdot 14 \cdot 15/(4 \cdot 84)$.

3. This simplifies to $R = 65/8$. Since $15^2 < 13^2 + 14^2$, the triangle is acute and its circumcenter is interior.

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 6 / 方法 6 [GE05-M06] (continued / 续)

例题解析

1. 半周长为二十一，海伦公式给出 $K=\sqrt{21\cdot 8\cdot 7\cdot 6}=84$。

2. 使用 $R=abc/(4K)=13\cdot 14\cdot 15/(4\cdot 84)$。

3. 化简为 $R=65/8$；由于 $15^2<13^2+14^2$，三角形为锐角，外心在内部。

A circumradius is not generally the distance from the centroid to a vertex.

外接圆半径一般不是重心到顶点的距离。

Practice / 配套练习: AIME-2000-II-Q12 (WB p. 430), AIME-2003-II-Q07 (WB p. 268), AIME-2017-I-Q04 (WB p. 443)

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 7 / 方法 7 [GE05-M07]

Incenter distances are half-angle right triangles

内心距离来自半角直角三角形

Procedure

1. Drop a radius to a side: $AI = r/\sin(A/2)$ and the adjacent tangent length is $r\cot(A/2)$.

2. The internal bisectors meet with $\angle BIC = 90^\circ + A/2$.

3. In an isosceles triangle, put the center on its symmetry axis and use one half-triangle.

4. For excircles use $r_a = K/(s-a)$; preserve external-center directions.

通用步骤

1. 向边作半径，得 $AI = r/\sin(A/2)$，邻接切线长为 $r\cot(A/2)$。

2. 内角平分线满足 $\angle BIC = 90^\circ + A/2$。

3. 等腰三角形中把内心放在对称轴上，使用半个三角形。

4. 旁切圆使用 $r_a = K/(s-a)$，并保留外部圆心方向。

> Example. In a triangle with inradius two and angle $A = 60^\circ$, find AI and $\angle BIC$.

> 例题：三角形内切圆半径二，$A = 60^\circ$，求 AI 与 $\angle BIC$。

Worked solution

1. At a contact point, the radius is perpendicular to the side and the angle at A is 30°.

2. Therefore $AI = 2/\sin 30^\circ = 4$.

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 7 / 方法 7 [GE05-M07] (continued / 续)

3. The bisector-angle identity gives $\angle BIC = 90^\circ + 30^\circ = 120^\circ$.

例题解析

1. 切点处半径垂直边，A 处夹角为 30°。

2. 所以 $AI = 2/\sin 30^\circ = 4$。

3. 平分线交角恒等式给出 $\angle BIC = 120^\circ$。

An angle between unoriented bisector lines may be supplementary to the angle between their inward rays.

无向角平分线的夹角，可能是向内射线交角的补角。

Three equal corner circles have centers homothetic to the vertices about the incenter, with ratio $(r - x)/r$. In an isosceles triangle with base a, equal sides b and altitude h, use $r = ah/(a + 2b)$, base exradius $ah/(2b - a)$ and other exradii h.

三个等半径顶角圆的圆心，是三顶点以内心为中心、比例 $(r - x)/r$ 的位似像。等腰三角形底边为 a、腰为 b、高为 h 时，可用 $r = ah/(a + 2b)$、底边旁切圆半径 $ah/(2b - a)$、另两旁切圆半径 h。

Practice / 配套练习: AIME-2007-II-Q15 (WB p. 343), AIME-2009-I-Q15 (WB p. 295), AIME-2009-II-Q10 (WB p. 371), AIME-2015-I-Q11 (WB p. 323), AIME-2018-I-Q13 (WB p. 324), AIME-2019-I-Q11 (WB p. 346), AIME-2026-II-Q08 (WB p. 318)

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 8 / 方法 8 [GE05-M08]

Euler’s center identity links inradius and circumradius

欧拉圆心恒等式联系内外接半径

Procedure

1. For incenter I and circumcenter O, use $OI^2 = R^2 - 2Rr$.

2. Combine a perpendicular-center condition with a right triangle involving a vertex on the circumcircle.

3. Recover a half-angle from $AI = r/\sin(A/2)$, then use $K = rs = bc\sin A/2$.

通用步骤

1. 对内心 I、外心 O，使用 $OI^2 = R^2 - 2Rr$。

2. 将圆心垂直条件与包含圆上顶点的直角三角形结合。

3. 由 $AI = r/\sin(A/2)$ 求半角，再用 $K = rs = bc\sin A/2$。

> Example. A triangle has $R = 5, r = 2$ and $IA \perp OI$. Find AI.

> 例题：三角形有 $R = 5, r = 2$ 且 $IA \perp OI$，求 AI。

Worked solution

1. Euler gives $OI^2 = 25 - 20 = 5$.

2. In right triangle OIA, $AI^2 = OA^2 - OI^2 = 25 - 5 = 20$.

3. Thus $AI = 2\sqrt{5}$ and $\sin(A/2) = 2/(2\sqrt{5}) = 1/\sqrt{5}$, consistent with a positive angle.

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 8 / 方法 8 [GE05-M08] (continued / 续)

例题解析

1. 欧拉恒等式给出 $OI^2 = 25 - 20 = 5$。

2. 直角三角形 OIA 中，$AI^2 = OA^2 - OI^2 = 20$。

3. 所以 $AI = 2\sqrt{5}$，且 $\sin(A/2) = 1/\sqrt{5}$，符合正角条件。

Write the right triangle with its actual right-angle vertex.

必须按实际直角顶点写勾股关系。

Practice / 配套练习: AIME-2024-II-Q10 (WB p. 317)

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 9 / 方法 9 [GE05-M09]

The bisector’s arc midpoint is equidistant from the incenter and endpoints

角平分线弧中点到内心及弧端点等距

Procedure

1. The internal bisector from A meets the circumcircle at the midpoint M of the arc BC not containing A.

2. Angle chasing gives $MB = MC = MI$ for the incenter I.

3. Use these equal distances with secant power, similar triangles or an antipodal point.

4. Keep the order of the bisector’s circle, side and incenter intersections explicit.

通用步骤

1. 从 A 出发的内角平分线交外接圆于不含 A 的弧 BC 中点 M。

2. 追角可得 $MB = MC = MI$，其中 I 为内心。

3. 将等距与割线点幂、相似三角形或对径点结合。

4. 明确角平分线上圆交点、边交点及内心的次序。

> Example. An internal bisector from C meets side AB at L, then its circumcircle at M. The order is C, I, L, M, with $LI = 1, LM = 2$. Find MC if $MA^2 = LM \cdot MC$.

> 例题：从 C 的内角平分线交 AB 于 L、再交外接圆于 M，点序为 C, I, L, M，且 $LI = 1, LM = 2$。已知 $MA^2 = LM \cdot MC$，求 MC。

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 9 / 方法 9 [GE05-M09] (continued / 续)

Worked solution

1. By the arc-midpoint lemma, $MA = MI$. The given order gives $MI = LI + LM = 3$.

2. The secant relation becomes $9 = 2MC$.

3. Thus $MC = 9/2$, and $CI = MC - MI = 3/2 > 0$ checks the stated order.

例题解析

1. 弧中点引理给出 $MA = MI$，由点序得 $MI = LI + LM = 3$。

2. 割线关系成为 $9 = 2MC$。

3. 所以 $MC = 9/2$，且 $CI = MC - MI = 3/2 > 0$，核实点序。

An antipode turns a cyclic right angle into a line relation; choose the correct arc midpoint first.

对径点可将圆内直角转为直线关系，但须先选对弧中点。

To prove $MA = MI$ in the version with bisector from C and arc midpoint M of AB, let half-angles at A, C be u, v. Both $\angle MAI$ and $\angle MIA$ equal $u + v$, making triangle AMI isosceles.

对从 C 平分至弧 AB 中点 M 的版本，设 A, C 半角为 u, v。则 $\angle MAI$ 与 $\angle MIA$ 均为 $u + v$，故 AMI 等腰，得到 $MA = MI$。

Practice / 配套练习: AIME-2012-II-Q15 (WB p. 297), AIME-2014-II-Q14 (WB p. 322), AIME-2016-I-Q06 (WB p. 316), AIME-2020-I-Q13 (WB p. 325)

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 10 / 方法 10 [GE05-M10]

Centroid, circumcenter and nine-point circle have exact coordinate roles

重心、外心与九点圆具有精确坐标角色

Procedure

1. The centroid is the average of the three vertex vectors and divides each median in ratio $2:1$.

2. Find a circumcenter from perpendicular-bisector equations, and use symmetry to identify it with the centroid only in special triangles.

3. The side midpoints and altitude feet lie on the nine-point circle; convert its arcs using inscribed angles.

4. When right-angle circles transfer directed angles, finish with a ray and supplement check.

通用步骤

1. 重心为三个顶点向量的平均，按 $2:1$ 分每条中线。

2. 外心由垂直平分线方程确定，只有特殊三角形才能凭对称性认定其与重心相同。

3. 三边中点及三条高的垂足共九点圆，可用圆周角换算弧。

4. 利用直角圆传递有向角后，最后仍须检验射线与补角分支。

> Example. Triangle vertices are $(0,0),(6,0),(0,6)$. Find its centroid and circumcenter.

> 例题：三角形顶点为 $(0,0),(6,0),(0,6)$，求重心与外心。

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 10 / 方法 10 [GE05-M10] (continued / 续)

Worked solution

1. The average vertex vector gives centroid $G = (2, 2)$.

2. The two axis legs have perpendicular bisectors $x = 3, y = 3$, so circumcenter $O = (3, 3)$.

3. They are distinct. As a right-triangle check, O is the midpoint of the hypotenuse.

例题解析

1. 顶点平均给出重心 $G = (2, 2)$。

2. 两条轴上直角边的垂直平分线为 $x = 3, y = 3$，所以外心 $O = (3, 3)$。

3. 两中心不同；直角三角形的斜边中点为外心，也核实了结果。

Directed-angle equality modulo 180° does not determine the ordinary angle without ray information.

模 180° 的有向角相等，不能在缺少射线信息时直接确定普通角。

Practice / 配套练习: AIME-2002-I-Q13 (WB p. 293), AIME-2011-II-Q13 (WB p. 395), AIME-2013-II-Q04 (WB p. 356), AIME-2021-II-Q14 (WB p. 307), AIME-2025-II-Q05 (WB p. 282)

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 11 / 方法 11 [GE05-M11]

Isogonal cevians give a sine-area product relation

等角劈线给出正弦面积乘积关系

Procedure

1. For cevians from one vertex, write each pair of adjacent subtriangle areas using a common pair of side lengths and angle sines.

2. Equal reflected angles exchange the sine factors, causing the product of opposite area ratios to simplify.

3. Use the remaining side ratios or area totals to solve, retaining which cevian is internal.

通用步骤

1. 对同顶点劈线，用共同侧边长度及夹角正弦表示相邻子三角形面积。

2. 关于角平分线对称的角交换正弦因子，使两组面积比乘积化简。

3. 再用剩余边比或面积总和求解，并保留内部劈线位置。

> Example. In triangle ABC, $AB = 2AC$. Rays AD, AE inside angle A satisfy $\angle BAD = \angle EAC$. Show $(BD/DC)(BE/EC) = 4$.

> 例题：三角形中 $AB = 2AC$，内部射线 AD, AE 满足 $\angle BAD = \angle EAC$。证明 $(BD/DC)(BE/EC) = 4$。

Worked solution

1. By common-altitude area ratios, $BD/DC = [ABD]/[ACD] = (AB/AC)\sin\angle BAD/\sin\angle DAC$.

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 11 / 方法 11 [GE05-M11] (continued / 续)

2. Similarly $BE/EC = (AB/AC)\sin\angle BAE/\sin\angle EAC$. Isogonality makes $\angle BAE = \angle DAC$.

3. Multiplication cancels both sine ratios, giving $(AB/AC)^2 = 4$.

例题解析

1. 由等高面积比，$BD/DC = (AB/AC)\sin\angle BAD/\sin\angle DAC$。

2. 同理 $BE/EC = (AB/AC)\sin\angle BAE/\sin\angle EAC$，而等角关系使 $\angle BAE = \angle DAC$。

3. 相乘后正弦比相消，得到 $(AB/AC)^2 = 4$。

Two equal-looking angles must be located at the same vertex before applying isogonality.

应用等角共轭前，应确认所比较的角位于同一顶点。

Practice / 配套练习: AIME-2005-II-Q14 (WB p. 274)

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 12 / 方法 12 [GE05-M12]

A midpoint perimeter bisector is parallel to the opposite angle bisector

过边中点的周长平分线平行于对角平分线

Procedure

1. Equal polygon perimeters after a cut mean the two original boundary routes between cut endpoints have equal length.

2. Through a side midpoint, construct the second endpoint using the difference of the adjacent side lengths.

3. Write its displacement as a scalar multiple of the sum of the two unit side directions at the opposite vertex.

4. The sum points along the internal angle bisector; convert the acute angle between two such lines with a supplement check.

通用步骤

1. 切开后的两多边形周长相等，等价于切点之间原边界两条路线等长。

2. 从边中点出发，利用相邻边长之差确定第二切点。

3. 将该切线向量写成对角顶点两边单位方向向量之和的倍数。

4. 单位向量和沿内角平分线，最后换算两条此类直线的锐夹角并检验补角。

> Example. Triangle ABC has $CA = 5, CB = 7$. Let M be the midpoint of AB, and $X \in CB$ with $CX = 1$. Prove MX bisects perimeter and is parallel to the C-bisector.

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 12 / 方法 12 [GE05-M12] (continued / 续)

> 例题：三角形中 $CA=5, CB=7$，M 为 AB 中点，$X \in CB$ 且 $CX=1$。证明 MX 平分周长并平行于 C 的角平分线。

Worked solution

1. Boundary routes $M-A-C-X$ and $M-B-X$ have lengths $AB/2+5+1$ and $AB/2+6$, hence equal.

2. With unit vectors u, v from C toward A, B, $M=(5u+7v)/2$ and $X=v$.

3. Thus $M-X=(5/2)(u+v)$, parallel to the internal bisector at C. Adding the common cut MX preserves equality of the two new perimeters.

例题解析

1. 边界路线 $M-A-C-X$ 与 $M-B-X$ 的长度分别为 $AB/2+5+1$ 与 $AB/2+6$，相等。

2. 以从 C 指向 A, B 的单位向量为 u, v，则 $M=(5u+7v)/2, X=v$。

3. 所以 $M-X=(5/2)(u+v)$，平行于 C 的内角平分线；两边都加入共同切段 MX 后，新周长仍相等。

Perimeter bisection and area bisection are different conditions.

平分周长与平分面积是不同条件。

Practice / 配套练习: AIME-2022-I-Q14 (WB p. 326)